NEOSTATS HACKATHON SUBMISSION

Automated Loan Refusal Prediction & Risk Intelligence

An end-to-end Machine Learning pipeline utilizing XGBoost, paired with an interactive 3-page Power BI executive suite for real-time credit decisioning and default prevention.

Validation AUC	OOT Test AUC	Refusal Threshold
0.9142	0.9375	0.75 Cutoff
Cross-validation split	Out-of-time stability	Conservative risk tuning

DATA ARCHITECTURE & HYGIENE

Relational Integration & Post-Decision Leakage Removal

Multi-Source Integration

Unified disparate transactional tables (`fact_applications`, `dim_client`) via primary keys `SK_ID_PREV` and `SK_ID_CURR`.

WHY: Consolidates demographics & history into 1 view.

Leakage Elimination

Purged post-decision variables like `CODE_REJECT_REASON` that only exist *after* loan outcome evaluation.

WHY: Prevents model from memorizing past manual decisions.

Domain Feature Engineering

Constructed debt stress proxies like `RATIO_APP_TO_INCOME` and categorical demographic bands (`AGE_BAND`, `INCOME_BAND`).

WHY: Captures applicant burden & non-linear risk segments.

ALGORITHM BENCHMARKING

Why XGBoost Was Selected as Production Engine

1. Superior Tabular & Non-Linear Modeling

Outperformed linear models (Logistic Regression) by naturally modeling complex feature interactions (income ratios vs. portfolio types) without manual transformation.

2. Native Imbalance & Missing Value Handling

Handles class imbalance (refusals vs. approvals) and missing application fields natively during tree construction without synthetic oversampling noise.

3. Outstanding Out-of-Time Generalization

Achieved **0.9375 AUC** on unseen temporal splits, proving zero overfitting and production readiness for future applications.

VALIDATION ROC-AUC

0.9142

OUT-OF-TIME (OOT) AUC

0.9375

Decision Cutoff: 0.75 Probability

Prioritizes high precision to eliminate costly false approvals.

REGULATORY TRANSPARENCY

Top Predictive Drivers of Loan Refusal

NAME_PORTFOLIO_POS
Point-of-Sale division indicator (#1 Risk Driver)
RANK #1 DRIVER

NFLAG_INSURED_ON_APPROVAL
Insurance attachment request flag
RANK #2 DRIVER

NAME_CLIENT_TYPE_Repeater
Returning vs new borrower history
RANK #3 DRIVER

RATIO_APP_TO_INCOME
Debt-to-income stress ratio
RANK #4 DRIVER



Regulatory Auditability

Tree-based feature importances ensure compliance with financial audit standards. Credit officers can clearly trace every refusal decision back to quantifiable applicant metrics rather than black-box outputs.

EXECUTIVE INTELLIGENCE TOOL

Interactive 3-Page Power BI Suite

PAGE 1

EDA & Portfolio Overview

Monitors total volume and baseline refusal rates (~21.7%) broken down by product lines (`NAME_PORTFOLIO_POS`) and client demographic bands.

WHAT & WHY: Global cross-filtering slicers.

PAGE 2

Model Performance

Displays validation ROC-AUC metrics (0.9142), 0.75 threshold callouts, and embeds the ROC curve graph for technical audit trust.

WHAT & WHY: Proves model discrimination.

PAGE 3

Predictions & Scoring

Operational data table displaying predicted probabilities and conditional risk bands (High, Medium, Low) for frontline credit officers.

WHAT & WHY: Actionable risk band slicers.